HW Plotting Requirements

1 Importance of Quality Plots

Having clear plots is crucial in science and engineering for several reasons, with three key reasons listed below.

1. **Quality plots enable effective communication** of complex data and concepts. Scientists and engineers often deal with intricate relationships, trends, and patterns within their data, and presenting these findings in a visual format helps convey information more efficiently and intuitively.

2. **Quality plots promote transparency and reproducibility** in scientific and engineering research. When plots are well-defined, properly labeled, and accompanied by appropriate axes, units, and legends, they enhance the clarity and accuracy of research presentations and publications.

3. **Quality plots showcase your attention to detail** as it involves accurately labeling axes, choosing appropriate scales, and ensuring that the plot is visually appealing and easily readable. These qualities are highly sought after by employers, as they indicate your ability to produce high-quality work with precision and professionalism.

2 Required Plotting Components

Plots will generally be graded as outlined in this text. All plots are required to be:

1. Have proper labels and units. This is typically shown as "label (SI unit)" on the x and y-axis. Proper scaling is also required, for example, reporting time in thousands of seconds rather than hours will not be accepted.

2. Be black and white friendly, both for printing and for those with visual limitations. Consider that almost 5% of the population is color blind; a number that increases with age and therefore seniority in companies. This is typically accomplished by using different line styles (solid, dashed, dotted) and appropriate colors. I highly recommend that students watch this video on the development of the Viridis color map `https://www.youtube.com/watch?v=xAoljeRJ3lU\&ab\_channel=Enthought`. It does a good job of laying out what goes into selecting colors in plots. The short version of this is to use the standard color maps in Python (Viridis) and MATLAB (Parula).

3. Figures should be appropriately sized, the font size should be approximately 11 pt and all fonts on the figure should be the same size.

4. Legends are required when needed. This is commonly the case when two lines appear on the same plot.

3 Example Plots

The following are examples of quality plots with key attributes annotated. In general, following these examples are what is required for this course.

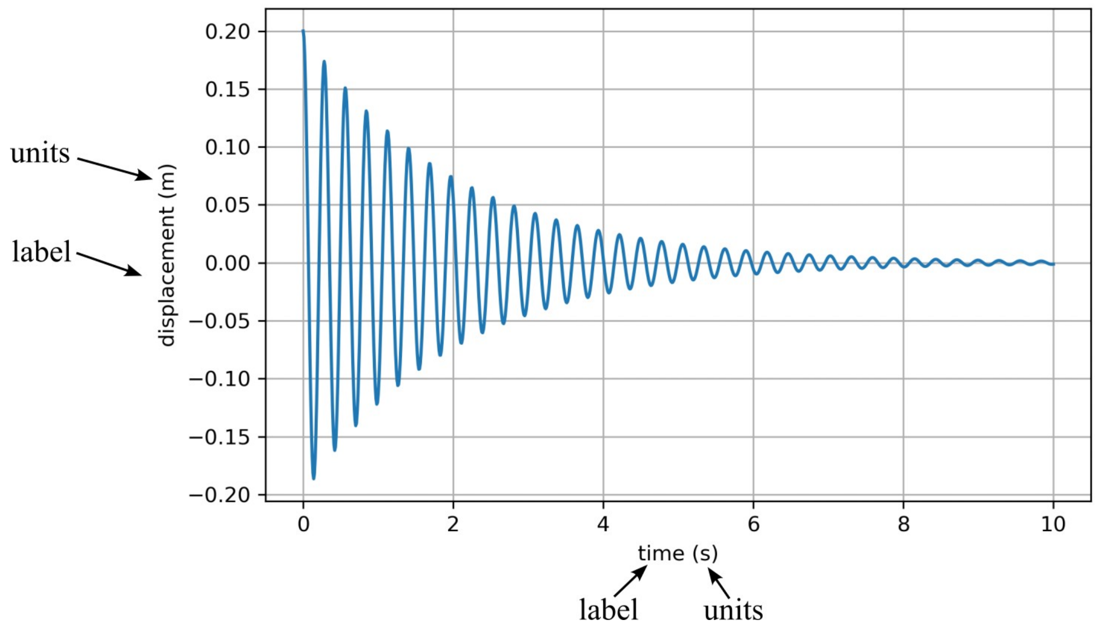

Figure 1: A figure showing what proper axis labels should look like.

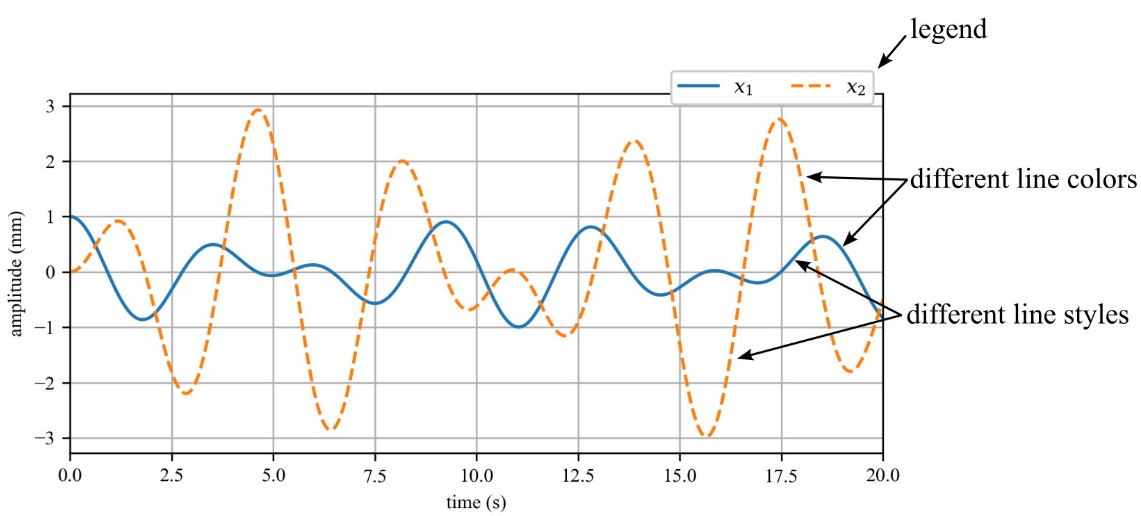

Figure 2: A figure showing what proper color and line selection should look like.

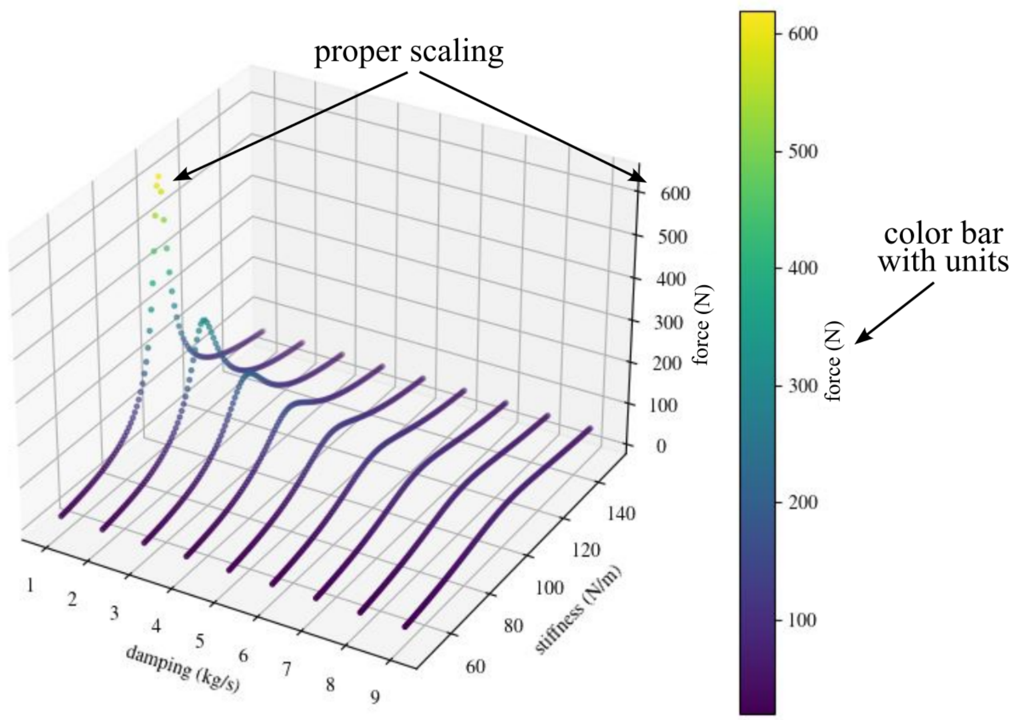

Figure 3: A figure showing what proper scaling should look like.